

CHIMERA PROMETHEUS

Fictional Build & Operations Manual

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Intended Use: Motion picture production only

■ **THIS IS A WORK OF FICTION.** All information—dimensions, pressures, temperatures, procedures, materials—is contrived for cinematic authenticity and has no basis in operational engineering safety. No part of this document may be used for actual construction, testing, or operation of rocket engines.

0. Preface and Safety Notice (for Production)

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1. Propellant Loading and Storage

1.1 Vehicle Tank Capacities (Single Prometheus Engine)

Propellant	Stage Tank Capacity	Density at Loading	Total Mass
LOX (liquid oxygen)	120 m³	1,141 kg/m³	136,920 kg
LCH4 (liquid methane)	82 m³	422 kg/m³	34,604 kg

Loading temperatures: LOX at 90 K, LCH4 at 111 K. Both propellants subcooled by 15 K below boiling point.

1.2 Propellant Conditioning Skid (Film Prop)

A fictional ground support module (“CryoForge Mark IX”) is described as a set piece: a six-pump skid with fractal-tube heat exchangers, actively cooled by liquid nitrogen circuits. Pumps are modelled visually as multi-stage centrifugal units with transparent housings.

1.3 Autogenous Pressurization Supply

Both tanks are pressurized by tapping the full-flow preburner exhaust. A fictional *electromagnetic gas compressor* (linear motor-driven piston) maintains tank pressure during high outflow.

2. Exact Mixture Schedules

2.1 Preburner Operation

Oxygen-rich preburner: Mixture ratio O/F = 5.8 (by mass). Turbine inlet temperature: 820 K.

Fuel-rich preburner: Mixture ratio O/F = 0.13 (very rich). Turbine inlet temperature: 1,050 K.

Both preburners operate at a chamber pressure of 48 MPa during nominal full-power mode. Combustion efficiency: 0.99 (ox-rich), 0.97 (fuel-rich).

2.2 Main Chamber Mixture Ratio

Annular RDRE main chamber designed for O/F = 3.8 (methane/oxygen). A 2% fuel-lean bias is introduced at the injector face (O/F = 3.88 overall), managed via the phase-locked injection waveform controller.

2.3 RDRE Injection Map (Cinematic Spec)

Four-wave mode ($N_w = 4$) at 35 MPa. Injector ring: 2,048 discrete elements with sinusoidal flow modulation superimposed on base flow.

Angular Sector	Base Mass Flux (kg/s·m²)	Amplitude Modulation	Phase Offset
0°–90°	850	+12%	–10°
90°–180°	850	–8%	+5°
180°–270°	850	+15%	0°
270°–360°	850	–10%	–5°

3. Startup Sequence and Mode Transitions

3.1 Pre-Start Ground Operations (T–2 hours)

- Cryo-cooling circuits active; bring superconducting elements below 80 K.
- Helium purge of all propellant lines and chamber.
- Digital twin synchronization – load nominal state vector and Bayesian priors.

3.2 Engine Spin-Up (T–10 seconds)

- Ox-rich preburner ignition via spark-torch igniters (6 units, 20 J each).
- Fuel-rich preburner ignition simultaneously.
- Motor-generators draw from ground capacitors (2 MJ flywheel bank) → 40% nominal RPM.
- Main chamber propellant valves cracked open; laser ignition (1064 nm, 50 mJ, 10 Hz, 12 locations).

3.3 Deflagration to Detonation Transition (T+3.5 s)

- Chamber builds to 15 MPa in deflagration mode.
- Phase-locked injection perturbation at ~2.4 kHz ($N_w=1$).
- Acoustic metamaterial liners damp transverse modes, amplify azimuthal modes.
- MHD plasma assist: pulsed 1 kA / 100 μs along detonation front path.
- T+3.8 s: detonation onset detected; controller shifts to full waveform map.

3.4 Full Rotating Detonation Operation

Wave stabilized at $N_w = 4$, chamber pressure 35 MPa. Turbopumps ramp to 100% nominal speed: fuel pump at 32,000 rpm; ox pump at 28,500 rpm. MHD generator extracts 150 kW for engine bus.

4. Pressurization Procedure

4.1 Tank Pressurization System

Autogenous pressurization supplemented by the **ElectroPulse** linear free-piston compressor (superconducting solenoid drive). Design parameters:

Stroke: 0.2 m | Bore: 0.15 m | Max frequency: 50 Hz
Compression ratio: 12:1 | Power per unit: 35 kW
Flight setpoints: LOX tank at 0.6 MPa, LCH4 tank at 0.5 MPa (±0.02 MPa trim).

4.2 Pre-Pressurization Sequence

- T-5 min: Ground helium pads tanks to 0.15 MPa.
- T-2 min: Autogenous pressurization begins; ElectroPulse units powered from ground.
- T-10 sec: Tanks at flight pressure; ground helium disconnects retract.

5. Injector Design and Hole Sizing

5.1 Main Annular Chamber Injector

Fractal-pattern concentric ring with 2,048 coaxial swirl elements. Each element: central LOX post (ø 1.2 mm) surrounded by annular methane gap (0.3 mm wide). Face curved to match chamber contour.

Material: Additively manufactured GRCop-42 with electroplated nickel and 10 µm iridium hot-face coating.

Injector pressure drop: ΔP = 4.2 MPa at full flow (acoustic decoupling).

Phase-locking actuation: Piezoelectric pintle per element, stroke 0.2 mm, up to 5 kHz.

5.2 Preburner Injectors (Impinging Triplets)

Preburner	Elements	Fuel Hole ø	LOX Hole ø
Fuel-rich	96	0.8 mm	0.5 mm
Ox-rich	120	0.4 mm	1.0 mm

6. Nozzle Dimensions and Plasma Actuation

6.1 Aerospike Plug Geometry

Annular aerospike with truncated plug, isentropic contoured expansion at exit area ratio 80:1 (geometric); plasma actuation extends effective area ratio to 120:1 in vacuum.

Parameter	Value
Inner plug length	2.4 m
Plug base diameter	1.2 m
Throat annular gap	0.08 m (±0.01 m adjustable)
Cowl lip radius	1.6 m
Geometric exit area ratio	80:1
Plasma-extended area ratio	120:1 (vacuum)

6.2 Superconducting Nozzle Coils

Three YBCO ring coils embedded in plug (77 K). Field up to 2.5 T at throat. Inner diameter 1.0 m, cross-section 0.15 × 0.15 m. Cooled by dedicated LCH4 loop.

6.3 MHD Electrodes

16 tungsten-copper electrode pairs at nozzle exit (water-cooled). Can extract power or generate a magnetic-pressure gradient for virtual nozzle extension.

7. Valve Timing and Actuation Sequences

7.1 Major Valves

All main valves are electro-pneumatic, actuated by 20 MPa helium (composite bottles). Response time < 50 ms.

7.2 Timing Chart (T=0 = ignition command)

Event	Time (s)	Action
Ox & fuel preburner igniters ON	T−2.0	Spark torch active
Preburner ox/fuel valves open	T−1.0	Linear ramp to 20% within 0.5 s
Preburner turbines spin-up	T−0.5	Motor assist on
Main chamber LOX & CH4 crack	T+0.5	5% initial flow
Laser ignition pulses start	T+0.6	10 Hz for 1 second
Deflagration established	T+1.5	Chamber pressure 10 MPa
RD transition sequence begin	T+3.0	Modulation + plasma assist
Detonation detected	T+3.8	Mode shift; wave number validated
Main valves ramp to 100%	T+4.5	Full propellant flow
MHD generator activation	T+5.0	150 kW power extraction commences

8. Turbopump Blade Geometry and Rotor Dynamics

8.1 Fuel Turbopump (Counter-Rotating Pair)

Parameter	Shaft 1 (Fuel-Rich Turbine)	Shaft 2 (Ox-Rich Turbine)
Inducer diameter	0.35 m	0.30 m
Impeller diameter	0.28 m	0.25 m
Blade count	7 main + 7 splitter	9 main + 9 splitter

Parameter	Shaft 1 (Fuel-Rich Turbine)	Shaft 2 (Ox-Rich Turbine)
Material	Ti-15Mo-3Nb-3Al + oxide	GRCop-42 + electroless Ni
Design RPM	32,000	28,500
Blade tip speed	470 m/s	—

Opposite rotation cancels gyroscopic torque. Radial flux-pinned superconducting magnetic bearings (YBCO Halbach array) + active magnetic thrust bearings on each shaft.

8.2 Motor-Generator Integration

Each shaft: 500 kW samarium-cobalt permanent magnet M/G. Provides startup torque; extracts or trims power during stable operation.

9. Material Heat-Treatment Recipes

9.1 GRCop-42 Chamber Liner

Process: Laser powder bed fusion, 30 µm layer.
Heat treat: Vacuum anneal 980°C / 1 hr → argon quench → age 480°C / 4 hr → furnace cool.
Yields fine Cu-Zr-O dispersion for high strength and thermal conductivity.

9.2 GRX-810 (ODS Superalloy) Structural Jacket

Process: Electron beam melting, 50 µm layer.
Post: HIP 1150°C / 150 MPa / 2 hr → solution 1200°C / 4 hr → oil quench → age 850°C/4h + 760°C/16h.

9.3 Diamond-Copper Heat Spreader Layer

2 mm layer, 60% diamond / 40% copper matrix, cold spray + hot press 600°C / 50 MPa.
Transverse thermal conductivity ≈ 800 W/m·K, smoothing detonation hot spots.

9.4 Transpiration Cooling Front Face

W-25Re by SLM, 30% porosity (20-50 µm pores).
Post: polymer infiltration → binder burnout → hydrogen sinter at 2000°C.

10. Test-Stand Operating Procedure (Film Version)

10.1 Stand Configuration

Engine mounted vertically into water-cooled flame deflector. Propellant run tanks: 50 m³ LOX, 35 m³ LCH4 (45 seconds at full power). DAQ: 5,000 channels at 100 kHz, fiber-optic, high-speed thermography.

10.2 Countdown and Firing

Time	Action
T-2 hr	Start cryocooling loops; bring superconductors to 77 K; begin tank chill-down.
T-30 min	Tank fill complete; ullage pressure stabilized by ElectroPulse compressors.
T-5 min	Activate magnetic bearings; spin sensor check; verify digital twin loaded.
T-60 sec	Helium purges; arm ignition circuits; verify valve latch positions.
T-10 sec	Preburners ignite; turbopumps motored to 20% speed.
T+0.5 s	Main chamber crack-open; laser ignition sequence begins.
T+1.5 s	Confirm steady deflagration; chamber pressure 10 MPa.
T+3.0 s	Engage RD transition sequence (modulation + plasma assist).
T+3.8 s	Detonation detected; wave number and stability validated.
T+5.0 s	Full thrust; MHD power extraction ON; engine self-powered.
T+45 s	Planned shutdown: reduce flow → deflagration → valve closure → N2 purge.

10.3 Post-Test

Borescope and thermal imaging of throat and nozzle. Download high-speed pressure data; update Bayesian digital twin. Anomalies trigger fictional safety review board.

11. New Speculative Concepts for the Film

11.1 Quantum Entanglement Sensor Web (QESW)

Network of nitrogen-vacancy centers in diamond chips embedded in the chamber wall, optically pumped and entangled — real-time magnetic field/temperature mapping with microsecond resolution. Visualized as a glowing 'quantum nervature' on the engine when active.

11.2 Negative-Index Metamaterial Layer for Radiative Cooling

Patterned photonic 'meta skin' (alternating Ag/Al2O3 nanolayers) exhibiting negative refractive index in mid-IR, bending thermal radiation for thermophotovoltaic harvesting.

11.3 Self-Healing Bioceramic Throat Insert

Calcium-phosphate porous ceramic infiltrated with low-melting alloy. Microcracks trigger alloy melting and re-solidification — depicted as a faint shimmer after test firing.

11.4 Temporal-Phase Detonation Control

Highly speculative plot device: A processor transmitting control signals via a fictional closed timelike curve resonator — allowing the controller to know the detonation state before it occurs. Introduced as a future upgrade that **malfunctions** and nearly destroys the engine.

Appendix A: Fictional Part Numbers and Suppliers

The following fictional parts are listed for set dressing purposes:

Part / Assembly	Part Number	Fictional Supplier
Magnetic Bearing Assembly	CP-MBA-88	Titan Dynamics
GRCop-42 Precursor Powder	Lot 7X-99	AeroMet Additive
ElectroPulse Linear Compressor	EP-4K	NovaTech
YBCO Superconducting Tape	SCS4050	Quantuum Conductors Ltd.

Appendix B: Visual References for Set Design

Engine Exterior

Towering metallic spike surrounded by an annular combustor ring with visible injector modulation actuators glowing blue-white (Cherenkov-like effect from fictional quantum sensors).

Plasma Nozzle Plume

Flared, multi-colored exhaust (amethyst, gold, sapphire) sculpted by magnetic fields, with visible MHD electrode arcs.

Turbopumps

Counter-rotating impellers visible through a transparent sapphire housing, bathed in coolant mist.

This Build & Operations Manual exists solely for the motion picture Chimera Prometheus. It is a work of imagination, designed to provide a plausible and visually stunning portrayal of a far-future rocket engine — a temperamental, living machine that pushes the boundaries of physics, and sometimes breaks them.

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